

Situational Pitch Efficacy

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01

Executive Summary



Project at a Glance



Built an interactive pitch analysis tool to evaluate how pitch type + location perform across specific game situations

Leverages pitch-level data (Statcast + advanced metrics) to estimate success probability by context (count, runners, handedness, etc.)

Enables users to:

- Identify optimal pitch-location combinations
- Compare outcomes across situations
- Visually isolate successful pitch patterns

02

Analytical Framework



Data Sources



Statcast Pitch-by-Pitch

2020-2024 Seasons



Fangraphs

Advanced Stats on how
pitch types perform on a
league-average basis



Brooks Baseball

Data on pitch performance,
batter-pitcher matchups

A pitch was classified as successful if it produced one of the following outcomes:

- Swinging strike (whiff)
- Called strike
- Weak contact, defined using Statcast contact quality:
 - Batted ball with launch speed-angle ≤ 3
 - Represents topped, under, or otherwise low-quality contact based on exit velocity and launch angle profiles from Statcast



Successful Pitch Definition

Methodology



Data Integration

Combined pitch-level data from Statcast, Brooks Baseball, and FanGraphs (2020–2024 seasons)
2. Data Cleaning & Standardization



Data Cleaning & Standardization

Filtered variables and harmonized pitch, outcome, and game-state fields across sources



Feature Engineering

Created binary game-state variables and defined custom pitch “success” outcomes (whiff, called strike, weak contact)



Situational Segmentation

Partitioned dataset into all combinations of count, outs, baserunners, and handedness



Outcome Aggregation

Computed empirical success rates for each pitch type and location within each game state



Interactive Output Generation

Structured results into an interface enabling filtering and visualization of realized pitch effectiveness

03

Analysis Viewing Interface



Situational Pitch Effectiveness

Pitcher Handedness

☒ R ☐ L

Batter Handedness

☐ R ☒ L

Outs

☐ 0 ☒ 1 ☐ 2

Balls

☐ 0 ☐ 1 ☒ 2 ☐ 3

Strikes

☐ 0 ☒ 1 ☐ 2

Bases Occupied

1st

2nd

3rd

Pitch Type

SL

Pitch Velocity

0

110

Pitch Spin Rate

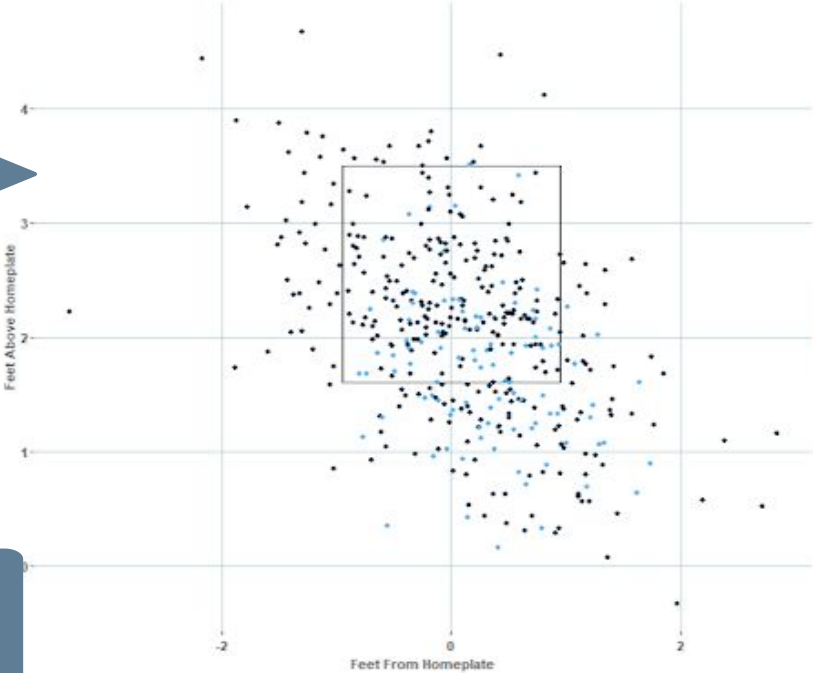
0

3,600

Success Rate: 0.241050119331742

Pitch Chart

Full Data



Game State Filters

All Pitches Displayed,
Successful Pitches
Highlighted

Pitcher Capability
Sliders

Pitch Charting Dashboard

04

Impact & Future Development



Applications



Live In-Game Pitch Calling

Adjustments to the dashboard can allow for in-game pitch calling* *(provided the rules allow for supplemental tools)

Self Scouting on Effective Location

Use pitch characteristic sliders to dial in the capabilities of the pitcher and assess corresponding effective locations

Expansion Opportunities

01

Conversion of pitch selector to checkbox format, allowing for full-arsenal view (separate graphics for each pitch type)



03

Modeling of expected gains and inclusion of game theory for upcoming pitch-location choices

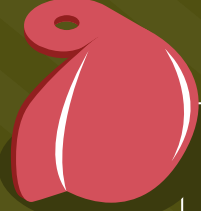


02

Inclusion of Batter profiles with layering of hot/cold zones

04

Incorporation of temporal context to capture pitch sequencing effects and prior pitch history within at-bat situations



Thank You

